Advanced Causal Inference

Lecture 09: Semiparametric Efficiency Theory (Parametric Model)

Chan Park
University of Illinois, Urbana-Champaign

This lecture note is partially based on the lecture notes/textbook of Andrea Rotnitzky

Outline

1. Review
2. Statistical Model & Parameter & Semiparametric Estimator
3. Cramer-Rao bound
4. Estimators

Three Estimators for the ATE

- Review

$$\mathrm{E}\{Y(1)\} = \mathrm{E}\left[\frac{AY}{e(X)}\right] \quad \text{(IPW)}$$

$$= \mathrm{E}\big[\mu(1,X)\big] \quad \text{(OR)}$$

$$= \mathrm{E}\left[\frac{A\{Y-\mu(1,X)\}}{e(X)} + \mu(1,X)\right] \quad \text{(DR)}$$

- Unbiased representation of $\mathrm{E}\{Y(1)\}$ under the model specification scenarios:

Unbiased Representation		e	
		Correctly specified	Mis-specified
μ	Correctly specified	(IPW), (OR), (DR)	(OR), (DR)
	Mis-specified	(IPW), (DR)	None

- This "double robustness" is often viewed as the primary reason for using the DR estimator

- However, there are other theoretical reasons why we prefer the DR estimator

Three Estimators for the ATE

- Review

$$\mathrm{E}\{Y(1)\} = \mathrm{E}\left[\frac{AY}{e(X)}\right] \quad \text{(IPW)}$$

$$= \mathrm{E}\big[\mu(1,X)\big] \quad \text{(OR)}$$

$$= \mathrm{E}\left[\frac{A\{Y-\mu(1,X)\}}{e(X)} + \mu(1,X)\right] \quad \text{(DR)}$$

- Unbiased representation of $\mathrm{E}\{Y(1)\}$ under the model specification scenarios:

Unbiased Representation		e	
		Correctly specified	Mis-specified
μ	Correctly specified	(IPW), (OR), (DR)	(OR), (DR)
	Mis-specified	(IPW), (DR)	None

- This "double robustness" is often viewed as the primary reason for using the DR estimator

- However, there are other theoretical reasons why we prefer the DR estimator

  $\Rightarrow$ Semiparametric Efficiency Theory

Semiparametric Efficiency Theory: A Personal Perspective

- This is my perspective on semiparametric efficiency theory and why it is essential for statisticians

1. **Provides theoretical justification for why intuitive statistical procedures are indeed well-founded**
   - For example, consider estimating the population mean $\mathrm{E}(Y)$
   - A natural estimator is the sample mean $\overline{Y}$
   - We know that $\overline{Y}$ is the MLE when $Y \sim N(\mu, \sigma^2)$, making it the optimal estimator in that case
   - But is $\overline{Y}$ still optimal if Y is not normally distributed?
   - Partial solution: Yes, if $Y \sim$ Poisson(μ), $\overline{Y}$ is the MLE.
   - This also holds for many other distributions, though each case must be assessed individually
   - But, this becomes not trivial when the distribution of Y can be "any" distributions
   - SET provides an answer to this question

2. **Offers a way (often the only one) to construct an asymp. normal estimator given a specified model**

- This may not be obvious initially. Consider the following example:
- If we assume $Y \sim N(\mu, \sigma^2)$, there are many possible estimators for μ, e.g., $\overline{Y}$ or Y_{median}
  note that both estimators are asymptotically normal
- If instead $Y \sim \text{Ber}(\mu)$, Y_{median} is no longer appropriate
- For a general model $Y \sim f \in \mathcal{M}$, which estimator is asymp. normal for $\mathrm{E}(Y)$?
- Of course, the answer is obvious: $\overline{Y}$ is asymp. normal thanks to the CLT
- But are there any other estimators that are asymp. normal for $\mathrm{E}(Y)$?
- Semiparametric theory provides guidance here

3. **Provides guidance on the proper use of nonparametric methods (ML, DNN, AI) when making $\sqrt{N}$-inference for common estimands, in both causal and non-causal settings**
   - Consider the OR estimator $\widehat{\tau}_{\mathsf{OR}} = \mathbb{P}\{\widehat{\mu}(1, X)\}$ for estimating $\mathrm{E}\{Y(1)\}$, where $\widehat{\mu}$ comes from a complex ML model
   - However, $\widehat{\tau}_{\mathsf{OR}}$ is an suboptimal estimator from the semiparametric efficiency theory perspective
   - From this viewpoint, many ML-based estimators are informative but not definitive, since they lack valid inference procedures

References for Semiparametric Efficiency Theory

- Tsiatis (2006)
  - Easy to read, provide a good introduction
- Newey (1990)
  - This paper is a GREAT introductory paper on SET
- Kennedy (2024)
  - This paper reinterprets SET considering recent advances in statistics and machine learning
- van der Vaart (1998, Chapter 25)
  - This is a very complete and rigorous but hard to read
- van der Vaart (2002)
  - This monograph expands upon the material presented in Chapter 25
- Bickel et al. (1998)
  - This book provides a rigorous semiparametric theory, but it is extremely difficult to read
- Kosorok (2007)
  - this book covers essentially the same material as van der Vaart (1998) at a slightly lower level
- van der Laan and Robins (2003)
  - mostly focuses on inference in semiparametric models with missing and coarsened at random data

Outline

1. Review
2. Statistical Model & Parameter & Semiparametric Estimator
3. Cramer-Rao bound
4. Estimators

General Setup

- Modern observational studies routinely collect **high-dimensional data** for each study unit
  - Typically including many baseline and time-dependent variables

- Scientific interest is usually in a **low-dimensional functional** of the data distribution F, e.g.

$$\beta(F)$$

- In practice, there is often **little or no knowledge** about the full distribution F

- The methods studied in this course are designed to address these analytic challenges
  - They yield valid inferences under semiparametric or nonparametric models
  - Such models require only minimal assumptions about parts of the data-generating law that are not of primary interest
  - Consequently, the methods are robust to misspecification of models for these secondary aspects of the law of the data

(Statistical) Model

- For the remainder of the lecture, "model" refers to a statistical model, not fashion supermodels

Parametric models for i.i.d data

Let $Z_1, \ldots, Z_N$ be N i.i.d. copies of a random structure Z, whose cdf F is assumed to belong to a family

$$\mathcal{F} = \{F_\theta \,|\, \theta \in \Theta\}$$

where $\Theta \subseteq \mathbb{R}^p$ is a subset of p-dimensional Euclidean space

Semiparametric models for i.i.d data

Let $Z_1, \ldots, Z_N$ be N i.i.d. copies of a random structure Z, whose cdf F is assumed to belong to a family

$$\mathcal{F} = \{F_\theta \,|\, \theta \in \Theta\}$$

where Θ is a "massive" set, i.e. it is **NOT** the subset of any Euclidean space

Parameter

Parameter

A parameter of $F_\theta \in \mathcal{F}$ is defined as $\beta(F_\theta) = \beta(\theta)$ for some function $\beta : \mathcal{F} \to \mathbb{R}^k$

- Even though semiparametric models do not require distributions to be dominated by a measure, say ν, we will assume they are dominated to avoid unnecessary technical complications
- We will use f to denote the density of F w.r.t. to the dominating measure ν
- in a slight abuse of notation, we will use to denote dx to denote $d\nu(x)$

Examples: Nonparametric model

- Assumes "nothing" about $\mathcal{F}$
- $\mathcal{F}$ is the collection of all probability distributions on a given sample space
- Θ is the collection of all probability distributions and each $\theta \in \Theta$ is a probability distribution
- We might be interested in estimating, for example, the mean of Z, i.e.

$$\beta(F) = \int z f(z) dz = \mathrm{E}(Z)$$

Examples: Symmetric distributions

- Suppose that Z is a scalar continuous r.v. with distribution F assumed to have a density f satisfying

$$f(z) = g(z - \beta^*)$$

for some unknown $\beta^* \in \mathbb{R}$ and some unknown function g which is symmetric around 0, i.e., $g(-u) = g(u)$

- Note that F is determined by the center of location β^* and the function g
- Therefore, the family can be indexed by $\theta = (\beta, g) \in \Theta = \mathbb{R} \otimes \mathcal{G}$
where $\mathcal{G} = \{$positive real valued functions on the reals that are symmetric around 0 and integrate to 1$\}$
- $\mathcal{F}$ is the collection of all probability distributions on a given sample space
- In this problem the parameter of interest is typically the center of location

$$\beta(F) = \text{the value } \beta^* \text{ such that } f(z) = g(z - \beta^*)$$

Examples: Conditional Mean Model

- Suppose that $Z = (Y, Z)$ where Y is a continuous response and X is a vector of covariates
- Consider a model $\mathrm{E}(Y|X) = g(X; \beta^*)$ where g is a known function (e.g., $g(X, \beta) = \beta^\intercal X$) and $\beta^* \in \mathbb{R}^k$ is unknown
- Let $\epsilon = Y - g(X; \beta^*)$
- Then β and the joint distribution of (ϵ, X) determine the joint distribution of (Y, X)

$$f_{YX}(y, x; \beta, \eta_1, \eta_2) = f_{Y|X}(y|x; \beta, \eta_1) f_X(x; \eta_2) = f_{\epsilon|X}(y - g(x; \beta)|x; \eta_1)\eta_2(x) = \eta_1(y - g(x; \beta), x)\eta_2(x)$$

where η_1 and η_2 are non-negative functions restricted only by

$$\int \eta_1(t, x)\, dt = 1 \ , \ \forall x \quad \text{and} \quad \int t\eta_1(t, x)\, dt = 0 \tag{1}$$

$$\int \eta_2(x)\, dx = 1 \tag{2}$$

- Thus the family $\mathcal{F}$ can be indexed by $\theta = (\beta, \eta_1, \eta_2) \in \mathbb{R}^k \otimes \mathcal{E}_1 \otimes \mathcal{E}_2$
  where $\mathcal{E}_1 = \{$all non-negative ft of (ϵ, x) satisfying (3)$\}$ and $\mathcal{E}_2 = \{$all non-negative ft of x satisfying (4)$\}$

Examples: Conditional Mean Model

- Suppose that $Z = (Y, Z)$ where Y is a continuous response and X is a vector of covariates
- Consider a model $\mathrm{E}(Y|X) = g(X; \beta^*)$ where g is a known function (e.g., $g(X, \beta) = \beta^\intercal X$) and $\beta^* \in \mathbb{R}^k$ is unknown
- Let $\epsilon = Y - g(X; \beta^*)$
- Then β and the joint distribution of (ϵ, X) determine the joint distribution of (Y, X)

$$f_{YX}(y, x; \beta, \eta_1, \eta_2) = f_{Y|X}(y|x; \beta, \eta_1) f_X(x; \eta_2) = f_{\epsilon|X}(y - g(x; \beta)|x; \eta_1)\eta_2(x) = \eta_1(y - g(x; \beta), x)\eta_2(x)$$

where η_1 and η_2 are non-negative functions restricted only by

$$\int \eta_1(\epsilon, x)\, d\epsilon = 1 \quad \text{and} \quad \int \epsilon\eta_1(\epsilon, x)\, dt = 0 \quad \text{for all } x \tag{3}$$

$$\int \eta_2(x)\, dx = 1 \tag{4}$$

- Thus the family $\mathcal{F}$ can be indexed by $\theta = (\beta, \eta_1, \eta_2) \in \mathbb{R}^k \otimes \mathcal{E}_1 \otimes \mathcal{E}_2$
  where $\mathcal{E}_1 = \{\text{all non-negative ft of } (\epsilon, x) \text{ satisfying (3)}\}$ and $\mathcal{E}_2 = \{\text{all non-negative ft of } x \text{ satisfying (4)}\}$
- In this problem the parameter of interest is the regression coefficient β^*:

$$\beta(F) = \text{the value } \beta^* \text{ such that } \epsilon = Y - g(X; \beta^*) \text{ satisfies (3)}$$

Semiparametric Estimators

- Loosely speaking, a semiparametric estimator of $\beta(F)$ is one which satisfies

$$\sqrt{N}\Big\{\widehat{\beta}_N - \beta(F)\Big\} \xrightarrow{D} N\big(0, \Sigma(F)\big) \ , \quad \forall F \in \mathcal{F}$$

- Consider $\mathcal{F}_1 = \{N(\mu, \sigma^2) \,|\, \mu \in \mathbb{R}, \sigma^2 \in (0, \infty)\}$. Then,

$$\sqrt{N}\big(\overline{Z} - \mu\big) \xrightarrow{D} N(0, \sigma^2) \quad \text{and} \quad \sqrt{N}\big(Z_{\text{median}} - \mu\big) \xrightarrow{D} N(0, 0.5\pi\sigma^2) \ , \quad \forall(\mu, \sigma^2)$$

- Both $\overline{Z}$ and Z_{median} are semiparametric estimators of μ under $\mathcal{F}_1$

- Consider $\mathcal{F}_2 = \{\text{Ber}(\mu) \,|\, \mu \in (0, 1)\}$. Then,

$$\sqrt{N}\big(\overline{Z} - \mu\big) \xrightarrow{D} N\big(0, \mu(1-\mu)\big) \ , \ \forall\mu \ , \quad \text{but} \quad \Pr(Z_{\text{median}} \in \{0, 1\}) = 1$$

- $\overline{Z}$ is still a semiparametric estimator of μ under $\mathcal{F}_2$ but Z_{median} is not a semiparametric estimator of μ under $\mathcal{F}_2$

- $\overline{Z}$ is still a semiparametric estimator of μ under the union model $\mathcal{F}_1 \cup \mathcal{F}_2$
  but Z_{median} is not a semiparametric estimator of μ under the union model $\mathcal{F}_1 \cup \mathcal{F}_2$

Semiparametric Estimators

- Questions:
  - For which functionals $\beta(F)$ can we hope to find semiparametric estimators?
  - When such estimators exist, how can we construct them?
  - How do we define an analogue of the Cramer-Rao variance bound in this setting?
  - For which functionals do "optimal" semiparametric estimators exist?
    i.e., estimators that achieve the smallest possible variance (the semiparametric Cramer-Rao bound)
  - If an "optimal" estimator exists, how can we construct it?

Outline

1 Review

2 Statistical Model & Parameter & Semiparametric Estimator

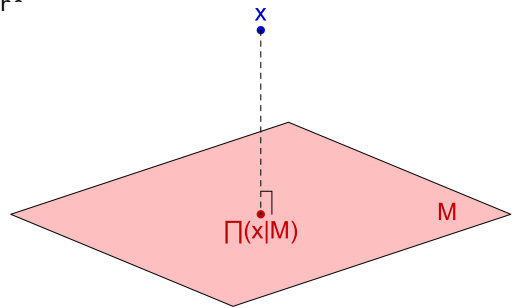
3 Cramer-Rao bound

4 Estimators

Projection

- Suppose $\boldsymbol{M} \in \mathbb{R}^k$ is a k-dimensional subspace spanned by the vectors $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$ with $k \geqslant p$, i.e.,

$$\boldsymbol{M} = \left\{ \sum_{j=1}^{p} a_j \boldsymbol{v}_j \,\middle|\, a_j \in \mathbb{R} \right\}$$



- Let $\Pi(\boldsymbol{x}|\boldsymbol{M})$ be the projection of $\boldsymbol{x}$ onto $\boldsymbol{M}$
- Since $\Pi(\boldsymbol{x}|\boldsymbol{M}) \in \boldsymbol{M}$, we have $\Pi(\boldsymbol{x}|\boldsymbol{M}) = \sum_{j=1}^{p} a_j \boldsymbol{v}_j$
- How to obtain $a_1, \ldots, a_p$?
- Residual $x - \Pi(\boldsymbol{x}|\boldsymbol{M})$ is orthogonal to the basis $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$, i.e.,

$$\left\langle x - \sum_{j=1}^{p} a_j \boldsymbol{v}_j, \boldsymbol{v}_i \right\rangle, \quad i = 1, \ldots, p \quad \Leftrightarrow \quad \sum_{j=1}^{p} a_j \langle \boldsymbol{v}_j, \boldsymbol{v}_i \rangle = \langle \boldsymbol{x}, \boldsymbol{v}_i \rangle \quad i = 1, \ldots, p$$

Projection

- Normal equation:

$$\sum_{j=1}^{p} a_j \langle \boldsymbol{v}_j, \boldsymbol{v}_i \rangle = \langle \boldsymbol{x}, \boldsymbol{v}_i \rangle \,, \quad i = 1, \ldots, p \quad \Leftrightarrow \quad \underbrace{\begin{bmatrix} \langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle & \langle \boldsymbol{v}_1, \boldsymbol{v}_2 \rangle & \cdots & \langle \boldsymbol{v}_1, \boldsymbol{v}_p \rangle \\ \langle \boldsymbol{v}_2, \boldsymbol{v}_1 \rangle & \langle \boldsymbol{v}_2, \boldsymbol{v}_2 \rangle & \cdots & \langle \boldsymbol{v}_2, \boldsymbol{v}_p \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \boldsymbol{v}_p, \boldsymbol{v}_1 \rangle & \langle \boldsymbol{v}_p, \boldsymbol{v}_2 \rangle & \cdots & \langle \boldsymbol{v}_p, \boldsymbol{v}_p \rangle \end{bmatrix}}_{\text{Gram matrix}} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix} = \begin{bmatrix} \langle \boldsymbol{x}, \boldsymbol{v}_1 \rangle \\ \langle \boldsymbol{x}, \boldsymbol{v}_2 \rangle \\ \vdots \\ \langle \boldsymbol{x}, \boldsymbol{v}_p \rangle \end{bmatrix}$$

$$\Leftrightarrow \quad (\boldsymbol{V}^\intercal \boldsymbol{V}) \boldsymbol{a} = \boldsymbol{V}^\intercal \boldsymbol{x}$$

- The OLS estimator is an projection estimator:

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \,, \qquad \boldsymbol{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_N \end{bmatrix} \in \mathbb{R}^N \,, \quad \boldsymbol{X} = \begin{bmatrix} | & \cdots & | \\ \boldsymbol{X}_1 & \cdots & \boldsymbol{X}_p \\ | & \cdots & | \end{bmatrix} \in \mathbb{R}^{N \times p}$$

$\widehat{\boldsymbol{\beta}}_{\mathsf{OLS}}$ = coefficients of $\{\boldsymbol{X}_1, \ldots, \boldsymbol{X}_p\}$ in $\Pi(\boldsymbol{Y} | \boldsymbol{X}_1, \ldots, \boldsymbol{X}_p)$

Score and Tangent Space

- Consider a parametric model

$$\mathcal{F} = \left\{ F(\cdot;\theta) \,\middle|\, \theta \in \Theta \subseteq \mathbb{R}^p \right\}$$

- Consider a log-likelihood function $\ell(z;\theta) = \log f(z;\theta)$

- The **score function** is defined as the derivative of the log-likelihood with respect to the parameter $\boldsymbol{S}(Z;\theta) = \left[S_1(Z;\theta), \ldots, S_p(Z;\theta) \right]^\intercal$ where

$$S_j(Z;\theta) = \nabla_j \log \ell(Z;\theta) = \frac{\nabla_j f(Z;\theta)}{f(Z;\theta)} \ , \quad j = 1, \ldots, p \qquad\qquad \nabla_j = \frac{\partial}{\partial \theta_j}$$

- The **tangent space for the model** $\mathcal{F}$ **at** θ is a subspace generated by the components of the score vector:

$$\Lambda(\theta) = \left\{ \boldsymbol{a}^\intercal \boldsymbol{S}(Z;\theta) \,\middle|\, \boldsymbol{a} \in \mathbb{R}^p \right\}$$

Fisher Information

- Note that the inner product of two random variables V and W is defined as $\langle V, W \rangle = \mathrm{E}(VW)$

- The gram matrix becomes

$$\begin{bmatrix} \langle \boldsymbol{v}_1, \boldsymbol{v}_1 \rangle & \langle \boldsymbol{v}_1, \boldsymbol{v}_2 \rangle & \cdots & \langle \boldsymbol{v}_1, \boldsymbol{v}_p \rangle \\ \langle \boldsymbol{v}_2, \boldsymbol{v}_1 \rangle & \langle \boldsymbol{v}_2, \boldsymbol{v}_2 \rangle & \cdots & \langle \boldsymbol{v}_2, \boldsymbol{v}_p \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \boldsymbol{v}_p, \boldsymbol{v}_1 \rangle & \langle \boldsymbol{v}_p, \boldsymbol{v}_2 \rangle & \cdots & \langle \boldsymbol{v}_p, \boldsymbol{v}_p \rangle \end{bmatrix}$$

$$= \begin{bmatrix} \mathrm{E}_\theta\big[S_1(Z;\theta)S_1(Z;\theta)\big] & \mathrm{E}_\theta\big[S_1(Z;\theta)S_2(Z;\theta)\big] & \cdots & \mathrm{E}_\theta\big[S_1(Z;\theta)S_p(Z;\theta)\big] \\ \mathrm{E}_\theta\big[S_2(Z;\theta)S_1(Z;\theta)\big] & \mathrm{E}_\theta\big[S_2(Z;\theta)S_2(Z;\theta)\big] & \cdots & \mathrm{E}_\theta\big[S_2(Z;\theta)S_p(Z;\theta)\big] \\ \vdots & \vdots & \ddots & \vdots \\ \mathrm{E}_\theta\big[S_p(Z;\theta)S_1(Z;\theta)\big] & \mathrm{E}_\theta\big[S_p(Z;\theta)S_2(Z;\theta)\big] & \cdots & \mathrm{E}_\theta\big[S_p(Z;\theta)S_p(Z;\theta)\big] \end{bmatrix}$$

$$= \mathrm{E}_\theta\big[\boldsymbol{S}(Z;\theta)\boldsymbol{S}(Z;\theta)^\intercal\big]$$

$$= I(\theta) \quad \text{aka the Fisher Information Matrix}$$

Efficiency Improvement

- Let $\mathcal{L}_2^0(\theta)$ be a collection of real-valued functions with zero mean and finite variance under F_θ:

$$\mathcal{L}_2^0(\theta) = \left\{ b(Z) \,\middle|\, \mathrm{E}_\theta[b(Z)] = 0,\ \mathrm{E}_\theta[g(Z)^2] < \infty \right\}$$

- If $I(\theta)$ is non-singular, the projection of $b(Z) \in \mathcal{L}_2^0(\theta)$ into the tangent space is

$$\Pi_\theta\big[b(Z) \,\big|\, \Lambda(\theta)\big] = \boldsymbol{a}^\intercal \boldsymbol{S}(Z;\theta)$$

where

$$\boldsymbol{a} = I(\theta)^{-1} \mathrm{E}_\theta\big[\boldsymbol{S}(Z;\theta) b(Z)\big]$$

- Therefore, we have

$$\Pi_\theta\big[b(Z) \,\big|\, \Lambda(\theta)\big] = \mathrm{E}_\theta\big[b(Z)\boldsymbol{S}(Z;\theta)^\intercal\big] I(\theta)^{-1} \boldsymbol{S}(Z;\theta)$$

Efficiency Improvement

- Fact: $\|b(Z)\|^2 \geqslant \|\Pi_\theta[b(Z)|\Lambda(\theta)]\|^2$
  - Similar to $\mathrm{Var}(Y) = \mathrm{Var}[\mathrm{E}(Y|X)] + \mathrm{E}[\mathrm{Var}(Y|X)] \geqslant \mathrm{Var}[\mathrm{E}(Y|X)]$

- Since $\mathrm{E}\{b(Z)\} = 0$, we have

$$\begin{aligned}
\mathrm{Var}_\theta\{b(Z)\} &= \|b(Z)\|^2 \\
&\geqslant \big\|\Pi_\theta\big[b(Z) \,\big|\, \Lambda(\theta)\big]\big\|^2 \\
&= \mathrm{Var}_\theta\Big[\mathrm{E}_\theta\big[b(Z)\boldsymbol{S}(Z;\theta)^\intercal\big] I(\theta)^{-1}\boldsymbol{S}(Z;\theta)\Big] \\
&= \mathrm{E}_\theta\big[b(Z)\boldsymbol{S}(Z;\theta)^\intercal\big] I(\theta)^{-1}\mathrm{Var}_\theta\big[\boldsymbol{S}(Z;\theta)\big] I(\theta)^{-1}\mathrm{E}_\theta\big[\boldsymbol{S}(Z;\theta)b(Z)\big] \\
&= \mathrm{E}_\theta\big[b(Z)\boldsymbol{S}(Z;\theta)^\intercal\big] I(\theta)^{-1}\mathrm{E}_\theta\big[\boldsymbol{S}(Z;\theta)b(Z)\big]
\end{aligned}$$

- For any mean zero statistic $b(Z)$, the projection of it onto the tangent space $\Lambda(\theta)$ has a smaller variance
  - Similar to Rao-Blackwellization: $\mathrm{E}\{\delta(X)\} = \theta$ and $T(X) =$ suff. stat. for θ. Then, $\mathrm{Var}\{\delta_{RB}(X)\} \leqslant \mathrm{Var}\{\delta(X)\}$

- A similar result holds for a vector statistic $b(Z)$

Efficiency Improvement

- Let $\boldsymbol{Z} = (Z_1, \ldots, Z_N)$ i.i.d.
- Let $\boldsymbol{S}(\boldsymbol{Z}, \theta) = \sum_{i=1}^N \boldsymbol{S}(Z_i, \theta)$ and $\Lambda(\theta)$ be the score and the tangent space at θ for a parametric model for $\boldsymbol{Z}$ indexed by $\theta \in \mathbb{R}^p$
- Suppose that $\widehat{\beta} := \widehat{\beta}(\boldsymbol{Z})$ is an unbiased estimator of an $\mathbb{R}^k$-valued parameter $\beta(\theta)$ in the parametric model

$$\mathrm{Var}_\theta\{b(Z)\} \geqslant \mathrm{E}_\theta\big[b(Z)\boldsymbol{S}(Z;\theta)^\intercal\big]\mathrm{Var}_\theta\big[\boldsymbol{S}(Z;\theta)\big]^{-1}\mathrm{E}_\theta\big[\boldsymbol{S}(Z;\theta)b(Z)\big]$$

$$\begin{aligned}
\Rightarrow \quad \mathrm{Var}_\theta(\widehat{\beta}) &\geqslant \mathrm{E}_\theta\big[\widehat{\beta}\boldsymbol{S}(\boldsymbol{Z};\theta)^\intercal\big]\mathrm{Var}_\theta\big[\boldsymbol{S}(\boldsymbol{Z};\theta)\big]^{-1}\mathrm{E}_\theta\big[\boldsymbol{S}(\boldsymbol{Z};\theta)\widehat{\beta}\big] \\
&= \mathrm{E}_\theta\big[\widehat{\beta}\boldsymbol{S}(\boldsymbol{Z};\theta)^\intercal\big]\mathrm{Var}_\theta\left[\sum_{i=1}^N \boldsymbol{S}(Z_i;\theta)\right]^{-1}\mathrm{E}_\theta\big[\boldsymbol{S}(\boldsymbol{Z};\theta)\widehat{\beta}\big] \\
&= \frac{1}{N}\mathrm{E}_\theta\big[\widehat{\beta}\boldsymbol{S}(\boldsymbol{Z};\theta)^\intercal\big]\mathrm{Var}_\theta\big[\boldsymbol{S}(Z;\theta)\big]^{-1}\mathrm{E}_\theta\big[\boldsymbol{S}(\boldsymbol{Z};\theta)\widehat{\beta}\big] \\
&= \frac{1}{N}\mathrm{E}_\theta\big[\widehat{\beta}\boldsymbol{S}(\boldsymbol{Z};\theta)^\intercal\big]I(\theta)^{-1}\mathrm{E}_\theta\big[\boldsymbol{S}(\boldsymbol{Z};\theta)\widehat{\beta}\big]
\end{aligned}$$

- However, this lower bound depends on the choice of an estimator;
  in other words, a different unbiased estimator $\widetilde{\beta}$ may lead to a different lower bound

Regular Parametric Model

Definition (Regular Parametric Model)

A parametric model is regular iff there exists a parameterization indexed by such that

1. Θ is an open subset of $\mathbb{R}^p$
2. For each $\theta \in \Theta$, the map $\theta \mapsto f(z;\theta)$ is continuous at θ
3. The map $\theta \mapsto \sqrt{f(z;\theta)}$ is **differentiable in quadratic mean** at all $\theta \in \Theta$
4. The information matrix $I(\theta)$ is non-singular for all $\theta \in \Theta$

- If the map $\theta \mapsto \sqrt{f(z;\theta)}$ is continuously differentiable and $I(\theta)$ is continuous, then 3 holds

Lemma

Let $\boldsymbol{Z} = (Z_1, \ldots, Z_N)$ i.i.d. and $T = t(\boldsymbol{Z})$ be a real-valued measurable ft.

Suppose that the parametric model $\mathcal{F}$ is regular; and that $\theta \mapsto \mathrm{E}_\theta(T^2)$ is continuous at θ^*. Then,

$$\frac{\partial}{\partial\theta}\mathrm{E}_\theta(T)\bigg|_{\theta=\theta^*} = \mathrm{E}_{\theta^*}\left\{T \cdot \boldsymbol{S}(\boldsymbol{Z};\theta^*)\right\}, \quad \boldsymbol{S}(\boldsymbol{Z};\theta) = \sum_{i=1}^{N} \boldsymbol{S}(Z_i;\theta)$$

Cramer-Rao Bound

- Lemma:

$$\frac{\partial}{\partial\theta}\mathrm{E}_\theta(T)\bigg|_{\theta=\theta*} = \mathrm{E}_{\theta*}\{T\cdot \boldsymbol{S}(\boldsymbol{Z};\theta^*)\}$$

- Applying the lemma for $\widehat{\beta}$ s.t. $\mathrm{E}(\widehat{\beta})=\beta$, the lower bound is expressed as

$$\mathrm{Var}_\theta(\widehat{\beta}) \geqslant \frac{1}{N}\mathrm{E}_\theta\big[\widehat{\beta}\boldsymbol{S}(\boldsymbol{Z};\theta)^\intercal\big] I(\theta)^{-1}\mathrm{E}_\theta\big[\boldsymbol{S}(\boldsymbol{Z};\theta)\widehat{\beta}\big] = \frac{1}{N}\underbrace{\frac{\partial\beta(\theta)}{\partial\theta^\intercal} I(\theta)^{-1}\frac{\partial\beta(\theta)}{\partial\theta}}_{C_{\mathcal{F}}(\theta)} = \frac{C_{\mathcal{F}}(\theta)}{N}$$

- The matrix

$$\underbrace{C_{\mathcal{F}}(\theta)}_{k\times k} = \underbrace{\frac{\partial\beta(\theta)}{\partial\theta^\intercal}}_{k\times p}\ \underbrace{I(\theta)^{-1}}_{p\times p}\ \underbrace{\frac{\partial\beta(\theta)}{\partial\theta}}_{p\times k}$$

  is called the **Cramer-Rao bound** (per unit of sample) for $\beta(\theta)$ in model $\mathcal{F}$ at the law F_θ

- Notice that this lower bound does not depend on which unbiased estimator we choose
  This bound applies to all unbiased estimators, making the bound universal and establishing it as a fundamental limit

Visual Summary

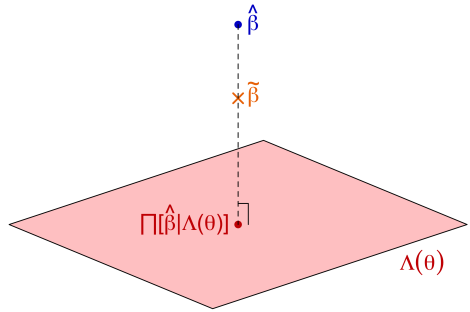
- Under a regular parametric model $F_\theta \in \mathcal{F}$, the projections of all unbiased estimators of $\beta(\theta)$ onto $\Lambda(\theta)$ are identical:

$$\forall\, \widehat{\beta}, \widetilde{\beta} \quad \text{such that} \quad \mathrm{E}_\theta[\widehat{\beta}] = \mathrm{E}_\theta[\widetilde{\beta}] = \beta(\theta)$$
$$\Rightarrow \quad \Pi_\theta\big[\widehat{\beta} \,\big|\, \Lambda(\theta)\big] = \Pi_\theta\big[\widetilde{\beta} \,\big|\, \Lambda(\theta)\big]$$

- The variance of unbiased estimators of $\beta(\theta)$ is no smaller than the Cramer Rao bound

- The result is not very useful; many estimators are biased

- Ridge, lasso, IPW, OR, and DR estimators, etc



$$\mathrm{E}_\theta(\widehat{\beta}) = \mathrm{E}_\theta(\widetilde{\beta}) = \beta(\theta)$$

Outline

1. Review
2. Statistical Model & Parameter & Semiparametric Estimator
3. Cramer-Rao bound
4. Estimators

Superefficient Estimators

- If $\widehat{\beta} \xrightarrow{P} \beta(\theta)$ at an appropriate rate, then $\mathrm{Var}_\theta\{\widehat{\beta} - \beta(\theta)\} \geqslant N^{-1} C_{\mathcal{F}}(\theta)$

- This is "almost true" but not quite true!

- One can construct "superefficient" estimators of $\beta(\theta)$ that have variance equal to the CR bound at most values of θ in the model,
  but at some values of θ have indeed variance smaller than the CR bound

$$\exists \widehat{\beta} \quad \text{s.t.} \quad \inf_\theta \left\{ \mathrm{Var}_\theta(\widehat{\beta}) - N^{-1} C_{\mathcal{F}}(\theta) \right\} < 0$$

- These estimators have excellent performance (as measured by MSE) at a θ^* at the expense of very poor performance in a neighborhood of θ^*

$$\mathrm{Var}_{\theta^*}(\widehat{\beta}) - N^{-1} C_{\mathcal{F}}(\theta^*) < 0 \quad \text{but} \quad \sup_{\theta \in \text{nbhd}(\theta^*)} \left\{ \mathrm{Var}_\theta(\widehat{\beta}) - N^{-1} C_{\mathcal{F}}(\theta) \right\} \gg 0$$

Hodges Estimator

- Example: Hodges estimator

- $Z_1, \ldots, Z_N \sim N(\beta, 1)$

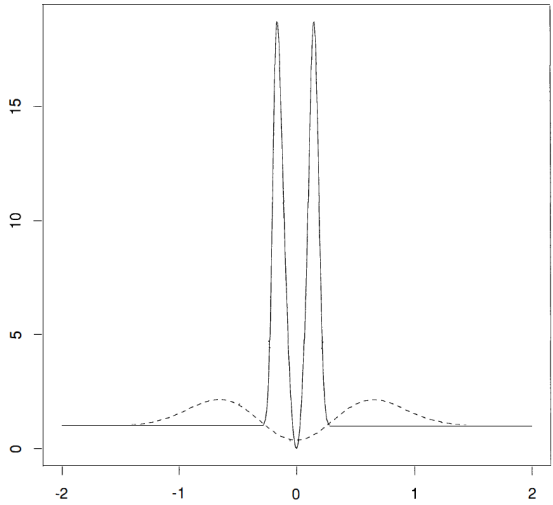
$$\widehat{\beta}_H = \begin{cases} \overline{Z} & \text{if } |\overline{Z}| > N^{-1/4} \\ 0 & \text{if } |\overline{Z}| \leqslant N^{-1/4} \end{cases}$$

- Then

$$\sqrt{N}(\widehat{\beta}_H - \beta) \overset{D(F_\beta)}{\to} \begin{cases} N(0,1) & \text{if } \beta \neq 0 \\ 0 & \text{if } \beta = 0 \end{cases}$$

- At $\beta = 0$, we have $N \cdot \mathrm{Var}(\widehat{\beta}_H) \to 0$, i.e., beats the CR bound!

Hodges Estimator



MSE of the Hodges Estimator based on $N = 10$ (dashed) and $N = 100$ (solid) observations from $N(\theta, 1)$

Regular Estimators

- Formulating a theory of asymptotic efficiency requires more than studying the convergence of estimators at each fixed θ
- We must analyze the distribution of $\widehat{\beta}$ across different parameter values as the sample size N increases
- From the construction of the Hodges estimator, the key parameter values lie in "shrinking neighborhoods" of a given parameter, which contract with N at an appropriate rate
- We consider a class of estimators that maintain asymptotic normality even when the parameter varies near a specific value

Regular Estimators

Definition (Regular Estimator)

Let $\mathcal{F} = \{F_\theta | \theta \in \Theta \subseteq \mathbb{R}^p\}$ be a parametric model and $\boldsymbol{Z} = (Z_1, \ldots, Z_N)$ N be i.i.d. observations from F_θ.
An estimator sequence $\widehat{\beta}_N = \widehat{\beta}(\boldsymbol{Z})$ is said to be a regular estimator of $\beta(\theta) \in \mathbb{R}^k$ at θ^* if there exists a law $G_{\theta*}$ such that

$$\sqrt{N}\{\widehat{\beta}_N - \beta(\theta^* + h/\sqrt{N})\} \overset{\theta^* + h/\sqrt{N}}{\to} G_{\theta*} \ , \quad \forall h$$

- Equivalently, let $\theta^*_N = \theta^* + h/\sqrt{N}$ and $\beta^*_N = \beta(\theta^* + h/\sqrt{N})$. Then, $\widehat{\beta}_N$ is regular iff

$$P_{\theta^*_N}\big[\sqrt{N}(\widehat{\beta}_N - \beta^*_N) \leqslant a\big] - P_{\theta*}\big[\sqrt{N}(\widehat{\beta}_N - \beta^*) \leqslant a\big] \to 0 \ , \quad \forall a, h$$

- A regular estimator's asymptotic distribution remains the same in shrinking neighborhoods of the true parameter value, i.e., the convergence of a regular estimator's distribution is locally uniform

- The class of regular estimators excludes the Hodges estimator, but it also excludes shrinkage estimators (e.g., James-Stein estimator)

Convolution Theorem

Theorem (Hajek's Convolution Theorem (van der Vaart, 1998, Theorem 8.8))

Let $\mathcal{F} = \{F_\theta | \theta \in \Theta \subseteq \mathbb{R}^p\}$ *be a* **regular** *parametric model, and* $\beta(\theta) \in \mathbb{R}^k$ *be a differentiable function of* θ*. If* $\widehat{\beta}_N$ *is a* **regular** *estimator of* $\beta(\theta)$ *at* θ^**, then*

$$\sqrt{N}\{\widehat{\beta}_N - \beta(\theta)\} \overset{D(F_\theta)}{\to} U + V$$

Here, $U \sim N_k(0, C_{\mathcal{F}}(\theta^*))$ *where* $C_{\mathcal{F}}(\theta^*)$ *is the CR bound for* $\beta(\theta^*)$ *in model* $\mathcal{F}$ *and* V *is independent of* U

- A regular estimator $\widehat{\beta}_N$ is most efficient when $V = 0$
- The CR bound is the smallest variance attainable by any regular estimator in a regular parametric model

Asymptotically Efficient Estimator

Definition (Asymptotically Efficient Estimator)

Let $\mathcal{F} = \{F_\theta | \theta \in \Theta \subseteq \mathbb{R}^p\}$ be a regular parametric model, and $\beta(\theta) \in \mathbb{R}^k$ be a differentiable function of θ.

- We call a regular estimator $\widehat{\beta}_N$ **locally** asymptotically efficient at θ if

$$\sqrt{N}\{\widehat{\beta}_N - \beta(\theta)\} \overset{D(F_\theta)}{\to} N_k(0, C_{\mathcal{F}}(\theta^*))$$

- We call a regular estimator $\widehat{\beta}_N$ **globally** asymptotically efficient if it is locally asymptotically efficient $\forall \theta \in \Theta$

- Can we always find globally asymptotically efficient estimators?

- Yes, if the map $\theta \mapsto f(x;\theta)$ satisfies one additional smoothness condition

- A globally efficient estimator is the maximum likelihood estimator

Asymptotically Linear Estimator & Influence Function

Definition (Asymptotically Linear Estimator)

Let F be a law and $\boldsymbol{Z} = (Z_1, \ldots, Z_N)$ be N i.i.d. observations from F. Let $\beta(F) \in \mathbb{R}^k$ be a parameter.

We call an estimator $\widehat{\beta}_N = \widehat{\beta}(\boldsymbol{Z})$ is asymptotically linear at F if

There exists a random vector $\varphi_F(Z)$ with $\mathrm{E}_F\{\varphi_F(Z)\} = 0$ and $\mathrm{Var}_F\{\varphi_F(Z)\} < \infty$ s.t.

$$\sqrt{N}\{\widehat{\beta}_N - \beta(F)\} = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \varphi_F(Z_i) + o_{P,F}(1) \tag{5}$$

where $o_{P,F}(1)$ denotes a sequence converging to 0 in probability under F

Definition (Influence Function)

The function $\varphi_F(Z)$ in (5) is referred to as the influence function (IF) of $\widehat{\beta}_N$ at F

- An asymptotically linear estimator has a unique IF
- The property of asymptotic linearity is NOT associated with a model
- It is a property at each, single law F, regardless if F is assumed to belong to a model $\mathcal{F}$

Asymptotically Linear Estimator & Influence Function

- Asymptotically linear estimators are important because many commonly used estimators are AL

- Furthermore, asymptotically efficient estimators are indeed asymptotically linear

Lemma 8.14 of van der Vaart (1998)

Let $\mathcal{F} = \{F_\theta | \theta \in \Theta \subseteq \mathbb{R}^p\}$ be a regular parametric model, and $\beta(\theta) \in \mathbb{R}^k$ be a differentiable function of θ.

Then, if $\widehat{\beta}_N$ is asymptotically efficient, it must be asymptotically linear

- Consequently, by focusing on the class of regular asymptotically linear estimators and selecting the one whose IF has the smallest variance, we do not lose efficiency.

Examples: OLS

- Suppose the law F generates (X, Y) satisfying that $Y = X^\intercal \beta + \epsilon$ with $\mathrm{E}(\epsilon|X) = 0$
- OLS estimator $\widehat{\beta}_N = \left[N^{-1} \sum_{i=1}^N X_i X_i^\intercal\right]^{-1} \left[N^{-1} \sum_{i=1}^N X_i Y_i\right]$
- One can show that $\widehat{\beta}_N$ is AL at F as

$$\sqrt{N}(\widehat{\beta}_N - \beta) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \mathrm{E}(XX^\intercal)^{-1} X \left(Y - X^\intercal \beta\right) + o_P(1)$$

with $\varphi_F(Y, X)$ written in red

Examples: Z-estimator

- Suppose that $U(Z;\beta)$ is a known $\mathbb{R}^k$-valued function s.t. there exists a unique β^* satisfying $\mathrm{E}_F[U(Z;\beta^*)] = 0$
- Z-estimator: $\widehat{\beta}_N$ satisfies $\frac{1}{N}\sum_{i=1}^N U(Z_i;\widehat{\beta}_N) = 0$
- Under certain regularity conditions (Chapter 5 of van der Vaart (1998)), one can show that $\widehat{\beta}_N$ is AL at F as

$$\sqrt{N}(\widehat{\beta}_N - \beta^*) = \frac{1}{\sqrt{N}}\sum_{i=1}^N \left[-\left.\frac{\partial \mathrm{E}_F[U(Z;\beta)]}{\partial \beta^\intercal}\right|_{\beta=\beta^*} \right]^{-1} U(Z_i;\beta^*) + o_P(1)$$

with $\varphi_F(Z)$ written in red

Examples: Maximum Likelihood Estimator

- MLE: $\widehat{\theta}_N$ satisfies $\widehat{\theta}_N = \arg\max_\theta \ell(\boldsymbol{Z}, \theta)$

- Under certain regularity conditions (Chapter 5 of van der Vaart (1998)), one can show that $\widehat{\theta}_N$ is AL at F as

$$\sqrt{N}(\widehat{\theta}_N - \theta) = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} \{ - I(\theta)^{-1} \boldsymbol{S}(Z_i, \theta) \} + o_P(1)$$

with $\varphi_F(Z)$ written in red

Regular and Asymptotically Linear Estimators

- An asymptotically linear estimator may not be regular; example: sample median
- The lemma below characterizes the conditions under which an asymptotically linear estimator is also regular

Lemma: Influence Function Equation

Let $\mathcal{F} = \{F_\theta | \theta \in \Theta \subseteq \mathbb{R}^p\}$ be a regular parametric model, and $\beta(\theta) \in \mathbb{R}^k$ be a differentiable function of θ. Suppose that $\widehat{\beta}_N$ is an asymptotically linear estimator of $\beta(\theta)$ at θ^* with an influence function $\varphi_{\theta^*}(Z)$.

Then, $\widehat{\beta}_N$ is regular at θ^* if and only if

$$\left.\frac{\partial \beta(\theta)}{\partial \theta^\intercal}\right|_{\theta=\theta^*} = \mathrm{E}_{\theta^*}\left\{\varphi_{\theta^*}(Z)\boldsymbol{S}(Z,\theta^*)^\intercal\right\} \qquad \text{(IFE-Para)}$$

where $\boldsymbol{S}(Z,\theta^*)$ is the score for θ at θ^*

- **RAL**: Regular and Asymptotically Linear
- OLS, Z-estimator, MLE satisfy (IFE-Para), meaning that they are RAL

Efficient Influence Function

- The Lemma has important implications
- The projections of all IFs of RAL estimators into the tangent space of the model coincide, which we call as the **Efficient Influence Function** (EIF):

$$\varphi_{\theta,\text{eff}}(Z) = \Pi_\theta\{\varphi_\theta(Z)|\Lambda(\theta)\}$$

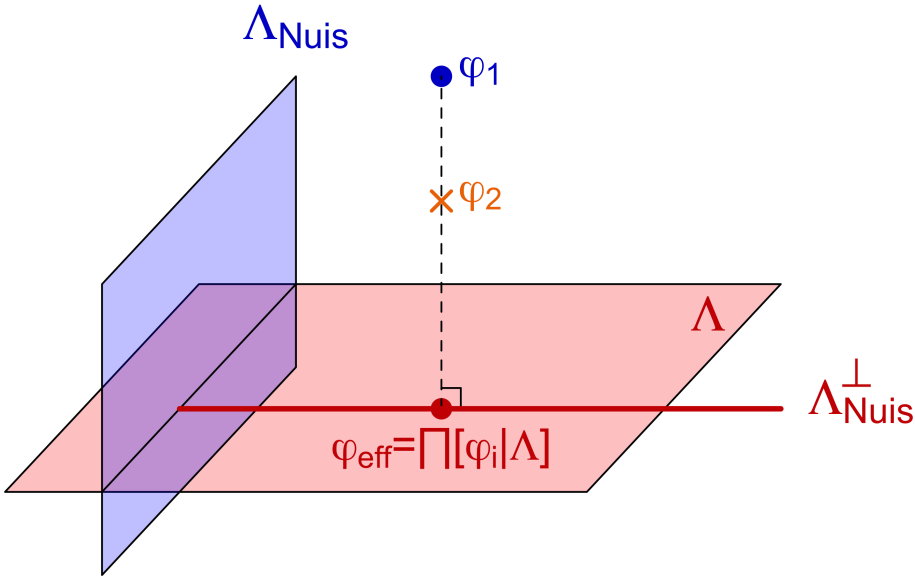
- Since the projection is unique, the EIF is unique
- The variance of the projection of any IF of a RAL estimator into the tangent space is equal to the CR bound

$$\text{Var}\big[\Pi_\theta\{\varphi_\theta(Z)|\Lambda(\theta)\}\big] = \text{Var}\big[\Pi_\theta\{\varphi_{\theta,\text{eff}}(Z)|\Lambda(\theta)\}\big] = C_{\mathcal{F}}(\theta)$$

- The set of all IFs of RAL estimators of $\beta(\theta)$ at F_θ is equal to

$$\big\{\varphi_\theta(Z)\big\} + \Lambda(\theta)^\perp = \big\{\varphi_{\theta,\text{eff}}(Z)\big\} \oplus \Lambda(\theta)^\perp$$

- The IFs of RAL estimators are orthogonal (i.e., uncorrelated) with the "nuisance tangent space"
- The covariance of any influence function with the score for β is equal to the identity


Λ_{Nuis}
φ_1
φ_2
Λ
$\Lambda_{Nuis}^{\perp}$
$\varphi_{eff}=\prod[\varphi_i|\Lambda]$

Example: OLS

- Consider the linear model:

$$Y = X^\intercal \beta + \epsilon \ , \quad \epsilon \sim N(0,1) \ , \quad X \sim F_X : \text{ known} \tag{6}$$

- Then, $\boldsymbol{S}(\beta, Y, X)$ is

$$\log f(Y, X, \beta) = -0.5 \log(2\pi) - \frac{(Y - X^\intercal \beta)^2}{2} \quad \Rightarrow \quad \boldsymbol{S}(\beta, Y, X) = X(Y - X^\intercal \beta)$$

- The IF of the OLS estimator, $\varphi_\beta(Y, X) = \mathrm{E}(XX^\intercal)^{-1} X(Y - X^\intercal \beta)$, is a linear combination of $\boldsymbol{S}(\beta, Y, X)$

- This implies $\varphi_\beta(Y, X) \in \text{span}\{\boldsymbol{S}(\beta, Y, X)\} = \Lambda(\beta)$

- Thus, $\varphi_{\beta,\text{eff}}(Y, X) = \Pi_\beta\{\varphi_\beta(Y, X) | \Lambda(\beta)\} = \varphi_\beta(Y, X)$

- The OLS estimator is the most efficient RAL estimator in the model (6)

Summary

- In a regular parametric model, the variance of any unbiased estimator attains the CR bound
- The CR bound does not necessarily apply to consistent estimators,
  as superefficient estimators can outperform the CR bound at specific parameter values
- For regular estimators, the CR bound holds
- Asymptotically linear estimators admit a linear representation with an IF, but not all AL estimators are regular
- If an IF satisfies (IFE-Para), the corresponding estimator is RAL
- Projecting the IF of a RAL estimator onto the tangent space yields the EIF
- The RAL estimator corresponding to the EIF attains the CR bound

References

Bickel, P. J., Klaassen, C. A., Ritov, Y., and Wellner, J. A. (1998). *Efficient and Adaptive Estimation for Semiparametric Models*. Springer, New York.

Kennedy, E. H. (2024). Semiparametric doubly robust targeted double machine learning: a review. *Handbook of statistical methods for precision medicine*, pages 207–236.

Kosorok, M. R. (2007). *Introduction to Empirical Processes and Semiparametric Inference*. Springer, New York.

Newey, W. K. (1990). Semiparametric efficiency bounds. *Journal of Applied Econometrics*, 5(2):99–135.

Tsiatis, A. A. (2006). *Semiparametric Theory and Missing Data*. Springer, New York.

van der Laan, M. J. and Robins, J. M. (2003). *Unified methods for censored longitudinal data and causality*. Springer, New York.

van der Vaart, A. W. (1998). *Asymptotic Statistics*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, New York.

van der Vaart, A. W. (2002). Semiparametric statistics. In *Lectures on Probability Theory and Statistics (Saint-Flour, 1999)*, pages 331–457. Springer.